
Notes for ‘Displacement, velocity and acceleration’

Important Ideas and Useful Facts:

- (i) **Displacement:** The *displacement* of an object or particle $x(t)$ at time t , relative to some starting point or origin, is typically modelled by position on the real line $\mathbb{R}$, so that the number 0 becomes a reference point, and we can make use of positive and negative directions.

The independent variable here is t , measured in some appropriate time units, typically seconds (but in applications may be minutes, hours, days or even years). Here the dependent variable is x (one of the few instances in which we conventionally choose x not to be the independent variable).

- (ii) **Velocity:** The *velocity* $v(t)$ of an object or particle at time t in appropriate units is the *instantaneous rate of change* of the displacement function $x(t)$ (though we need the notion of *derivative*, introduced later, to make this precise).

The velocity may be positive, negative or zero. The magnitude of velocity is known as *speed* (ignoring sign or direction).

For example, the speedometer of a vehicle, like a car or truck, in principle, displays the instantaneous speed at any given moment.

- (iii) **Acceleration:** The *acceleration* $a(t)$ of an object or particle at time t in appropriate units is the *instantaneous rate of change* of the velocity function $v(t)$ (though again we need the notion of *derivative*, introduced later, to make this precise).

The acceleration also may be positive, negative or zero.

The word *deceleration* is often used in applications when the acceleration happens to be negative and is quoted as a positive number (and it is then understood that the acceleration is the negative of this number).

The deceleration of an object in free-fall near the earth’s surface is approximately constant, with the value 9.8 metres per second per second, that is

$$a = a(t) = -9.8 \text{ m/sec}^2 .$$

This fact leads naturally to linear functions for the velocity function $v = v(t)$ and a quadratic function for the displacement function $x = x(t)$ (though the precise reason for these claims won’t become apparent until near the end of this course).

Examples:

1. A missile is launched directly into the air from a platform with displacement function $x = x(t)$ metres above the ground, at time t seconds after launching, approximated by the formula

$$x = x(t) = -5t^2 + 250t + 15 .$$

Estimate the following:

- (a) the height of the missile at the moment it was launched;
- (b) the height of the missile 10 seconds after it was launched;
- (c) the height of the missile 20 seconds after it was launched;
- (d) the average velocity of the missile over the first 10 seconds;
- (e) the average velocity of the missile over the period 10 to 20 seconds after it was launched;
- (f) the maximum height achieved by the missile above the ground and when this occurs.

Solutions:

- (a) The height of the missile at the moment it was launched is

$$x(0) = 0 + 0 + 15 = 15 \text{ m} .$$

- (b) The height of the missile 10 seconds after it was launched is

$$x(10) = -5(10^2) + 250(10) + 15 = 2,015 \text{ m} .$$

- (c) The height of the missile 20 seconds after it was launched is

$$x(20) = -5(20^2) + 250(20) + 15 = 3,015 \text{ m} .$$

- (d) Using the answers to parts (a) and (b), the average velocity of the missile of the first 10 seconds is

$$\frac{2,015 - 15}{10} = 200 \text{ m/sec}$$

(equivalently 720 kph).

- (e) Using the answers to parts (b) and (c), the average velocity of the missile over the period 10 to 20 seconds after it was launched is

$$\frac{3,015 - 2,015}{10} = 100 \text{ m/sec}$$

(equivalently 360 kph).

- (f) To find the maximum height achieved by the missile, we complete the square as follows:

$$x(t) = -5t^2 + 250t + 15 = -5(t^2 - 50t + 625 - 625) + 15 = 3140 - 5(t - 25)^2 ,$$

so that the maximum height is 3,140 m, which occurs $t = 25$ seconds after launching.

2. An arrow is shot upwards from the surface of the moon. Its height $x = x(t)$ metres above the surface after t seconds is approximated by the formula

$$x = x(t) = 2 + 40t - 0.8t^2 .$$

Its velocity and acceleration after t seconds (and before it lands) are approximated by the formulae

$$v = v(t) = 40 - 1.6t \quad \text{and} \quad a = a(t) = -1.6 .$$

Estimate the following:

- the height of the arrow 20 seconds after it was shot upwards;
- the height of the arrow 40 seconds after it was shot upwards;
- the average velocity of the arrow over the first 20 seconds;
- the average velocity of the arrow over the period 20 to 40 seconds after it was shot upwards;
- the maximum height reached by the arrow and when this occurs;
- the times at which the velocities of the arrow are 20 m/sec and -20 m/sec respectively.

Solutions: (a) The height of the arrow 20 seconds after it was shot upwards is

$$x(20) = 2 + (40)(20) - (0.8)(20^2) = 482 \text{ m} .$$

- (b) The height of the arrow 40 seconds after it was shot upwards is

$$x(40) = 2 + (40)(40) - (0.8)(40^2) = 322 \text{ m} .$$

- (c) The initial height of the arrow is $x(0) = 2$ m, so, using the answer to part (a), the average velocity of the arrow over the first 20 seconds is

$$\frac{482 - 2}{20} = 24 \text{ m/sec}$$

(which is about 86 kph).

- (d) Using the answers to parts (a) and (b), the average velocity of the arrow over the period 20 to 40 seconds after it was shot upwards is

$$\frac{322 - 482}{20} = \frac{-160}{20} = -8 \text{ m/sec}$$

(which is about 30 kph directed back towards the moon's surface).

- (e) To find the maximum height reached by the arrow, we complete the square as follows:

$$x(t) = 2 + 40t - 0.8t^2 = 2 - 0.8(t^2 - 50t + 625 - 625) = 502 - 0.8(t - 25)^2 ,$$

so that the maximum height is 502 m, which occurs $t = 25$ seconds after the arrow was shot upwards.

- (f) Observe that $v = v(t) = 20$ when $20 = 40 - 1.6t$, which is after

$$t = \frac{20}{1.6} = 12.5 \text{ seconds} ,$$

and $v = v(t) = -20$ when $-20 = 40 - 1.6t$, which is after

$$t = \frac{60}{1.6} = 37.5 \text{ seconds} .$$